UNIVERSITY OF ASIA PACIFIC

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 406

Microprocessor and Embedded System Laboratory

EXPERIMENT NO. 09

DESIGN AND DEVELOPMENT OF A PORTABLE MONITOR

Individual Technical Report

SUBMITTED BY	SUBMITTED TO
Md. Mosheur Rahman	Mr. Sakhawat Hossen Rakib
ID: 22208169	Assistant Professor
Section: B-2	Department of EEE
Department of EEE	University of Asia Pacific

July 2026

Abstract

This report presents the design, hardware integration, enclosure fabrication, assembly, and validation of a low-cost portable monitor prototype. The author personally completed the core hardware work, including integration of the LCD panel with a compatible display controller board, arrangement of the internal connections, design of the external shape, fabrication of the PVC-board enclosure, final assembly, troubleshooting, and functional testing. A small amount of occasional assistance was received from peers during component sourcing and troubleshooting. The prototype successfully displayed output from a computer source and provided accessible on-screen-display control buttons and external input/output connections. The recorded prototype cost was BDT 4,840. The project demonstrates a practical method of converting a compatible LCD panel and controller board into a functional portable display using locally available materials.

Project Contribution Statement

The practical implementation described in this report was led and carried out by the author. The hardware integration, enclosure shape and design, PVC-board fabrication, component mounting, assembly, troubleshooting, and final output testing were personally completed by the author. Several individuals provided occasional assistance during procurement, handling, and troubleshooting. The present report has been technically reviewed and rewritten to reflect the actual work performed.

1. Introduction

A portable monitor is a compact external display that can be connected to a laptop, desktop computer, media device, or embedded platform. Commercial portable monitors offer convenient secondary-screen functionality but may be expensive or difficult to repair. This project explored whether a functional portable monitor could be built at lower cost by combining a reusable LCD panel, a compatible controller board, an external power adapter, and a custom enclosure fabricated from PVC board.

The project focused on practical hardware integration rather than custom embedded firmware or backend software. The display controller board performs the required video processing, while its built-in on-screen-display button board provides power, menu, volume, channel, and input-selection controls.

1.1 Objectives

- To design and build a functional portable monitor prototype using locally available components.
- To interface an LCD panel with a compatible multi-input display controller board.
- To design and fabricate a low-cost protective enclosure using PVC board.
- To arrange the controller ports and OSD buttons so that they remain accessible after assembly.
- To test power-on operation, video output, image stability, control-button access, and structural support.
- To estimate the prototype cost and identify limitations for future improvement.

1.2 Scope

The scope includes hardware selection, physical integration, enclosure fabrication, assembly, basic functional testing, and cost analysis. The project does not claim a custom firmware design, cloud service, battery-management system, formal product certification, or mass-production readiness. The exact native resolution and detailed electrical specifications of the reused LCD panel should be verified from the panel label or manufacturer datasheet before commercial use.

2. Design Requirements

ID	Requirement	Design Response
R1	Functional display output	The monitor must display a stable image from a compatible source device.
R2	Low prototype cost	The design should use affordable, locally available components and materials.
R3	Portable form	The complete unit should be compact enough to move and use as a secondary display.
R4	Accessible controls	Power and OSD buttons must remain accessible from the front of the enclosure.
R5	Accessible ports	The enclosure must provide openings for the controller-board connectors.
R6	Mechanical support	The screen and controller must remain securely mounted during normal handling.

3. Components and Bill of Materials

The following costs were recorded during prototype development. Transportation is listed separately because it is not a physical component but contributed to the total prototype expenditure.

SL	Item	Quantity	Unit Cost (BDT)	Total (BDT)
1	LCD display panel	1	1,650	1,650
2	T.R67.03 display controller board	1	1,700	1,700
3	External DC adapter	1	100	100
4	PVC-board enclosure and fabrication materials	1 set	500	500
5	Cables and connectors	1 set	90	90
6	Transportation / conveyance	1 lot	800	800
	Total prototype expenditure			**4,840**

4. System Design and Working Principle

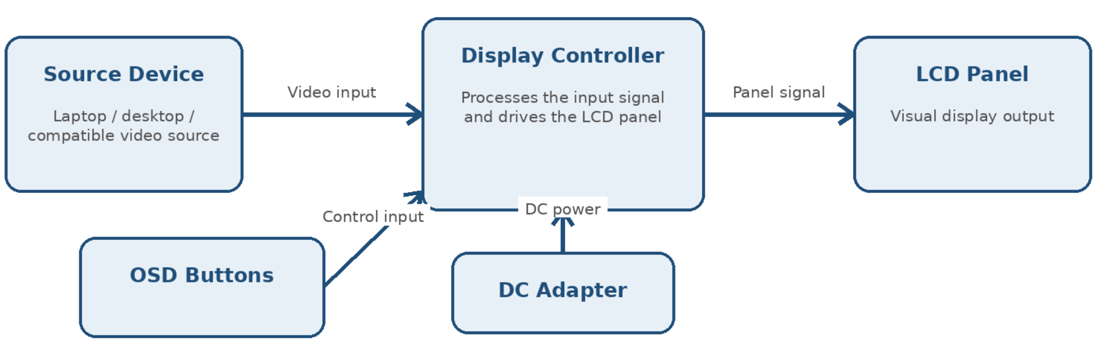

Figure 1: Functional block diagram of the portable monitor prototype.

A compatible source device sends a video signal to the display controller board. The controller board processes the signal and generates the electrical interface required by the LCD panel. An external DC adapter powers the controller and display circuitry. The OSD button board provides user control over power, menu navigation, volume or parameter adjustment, and input selection. The enclosure protects the components and exposes the required buttons and connectors.

4.1 Hardware Integration

- The LCD panel was paired with a compatible controller board after checking connector compatibility and successful display operation.
- The controller board and associated button board were positioned inside the lower enclosure section.
- Signal and power cables were arranged to reduce strain on the panel connector and to avoid interference with the support structure.
- Openings were cut in the enclosure so that the controller-board ports and front control buttons remained usable.
- The unit was powered and connected to a computer source for output verification.

4.2 OSD Control Interface

Control	Function
Power	Turns the monitor on or off.
CH+	Moves upward through available menu options.
CH-	Moves downward through available menu options.
VOL+	Increases the selected value or volume setting.
VOL-	Decreases the selected value or volume setting.
Menu	Opens the OSD menu or confirms a selection.
AV/TV	Selects or cycles through available input sources.

5. Mechanical and Enclosure Design

The external shape and enclosure were personally designed and fabricated by the author using PVC board. PVC board was selected because it was inexpensive, locally available, easy to cut, lightweight, and sufficiently rigid for a proof-of-concept enclosure. The approximate front dimensions recorded during development were about 28 cm in width and 18 cm in height. Because the rear support and lower controller compartment have different depths, a single uniform depth value was not used.

The enclosure consists of a front bezel around the LCD panel, a lower compartment for the controller and control board, a rear panel, and a central support structure. The rear/base section includes cut-outs aligned with the available controller-board connectors. The front lower section includes openings for the power indicator or receiver area and the OSD button board.

Figure 2: Front view of the assembled enclosure.

Figure 3: Rear view showing the support and exposed ports.

Figure 4: Side view of the PVC-board support structure.

5.1 Enclosure Design Features

- A custom front frame holds the display panel in position.
- A lower enclosure section houses the controller board and related wiring.
- A rear support helps maintain the viewing angle and mechanical stability.
- Port cut-outs allow cables to be connected without opening the enclosure.
- Front cut-outs provide access to the OSD controls.
- The design can be repaired or modified using common hand tools.

6. Assembly Procedure

1. Inspect the LCD panel and identify its connector type, supply requirements, and physical dimensions.
2. Select a compatible display controller board and verify basic operation before enclosure fabrication.
3. Measure the panel, controller board, connector positions, and OSD button-board location.
4. Prepare the enclosure layout and determine the front opening, lower compartment, rear panel, and support position.
5. Cut the PVC-board pieces and create openings for the display, control buttons, power connection, and available input/output ports.
6. Mount the LCD panel securely in the front frame without applying excessive pressure to the glass or flex cables.
7. Install the controller board and OSD button board inside the lower enclosure section.
8. Connect the panel cable, button cable, power cable, and required video-input cable; then arrange the wiring to minimize strain.
9. Join the enclosure parts, install the rear support, and check that the monitor stands securely.
10. Apply power, connect a source device, test the image and controls, and correct any connection or fitting problems.

6.1 Development Timeline

Period	Activity	Issue	Resolution
1-7 June 2026	Component procurement and compatibility search	Difficulty finding a controller compatible with the LCD panel.	A suitable controller board was obtained and tested with the panel.
8-14 June 2026	Initial enclosure planning and fabrication approach	The available 3D-printing area was smaller than the required enclosure dimensions.	The enclosure was redesigned for manual fabrication using PVC board.
15-20 June 2026	Hardware installation, assembly, and troubleshooting	Connector alignment, internal spacing, and stable mounting required repeated adjustment.	The internal layout and cut-outs were refined through multiple fitting and testing stages.

The first successful power-on was achieved on 18 June 2026, followed by final fitting and additional output testing.

7. Testing and Validation

The prototype was tested as a proof-of-concept device. The tests focused on observable functionality and mechanical usability. Long-duration thermal testing, electrical certification, colour calibration, and full port-by-port validation were outside the scope of this prototype stage.

ID	Test	Expected Result	Observed Result	Status
T01	Power-on test	The monitor powers on without visible electrical fault.	The unit powered on successfully.	PASS
T02	Video-input detection	The controller detects a connected computer source.	The source signal was detected.	PASS
T03	Display-output test	A stable and readable image appears on the LCD panel.	The Windows lock screen and desktop were displayed.	PASS
T04	OSD accessibility	The physical control buttons remain accessible after assembly.	The front controls were accessible and usable.	PASS
T05	Connector accessibility	Required cables can be connected through enclosure cut-outs.	The exposed rear ports accepted the test connections.	PASS
T06	Mechanical stability	The panel and support remain stable during normal demonstration.	The assembled prototype remained upright and supported.	PASS
T07	Extended thermal test	The unit operates safely over a long continuous period.	Not formally measured during this phase.	NOT TESTED

7.1 Demonstrated Output

Figure 5: Successful display of the Windows lock screen.

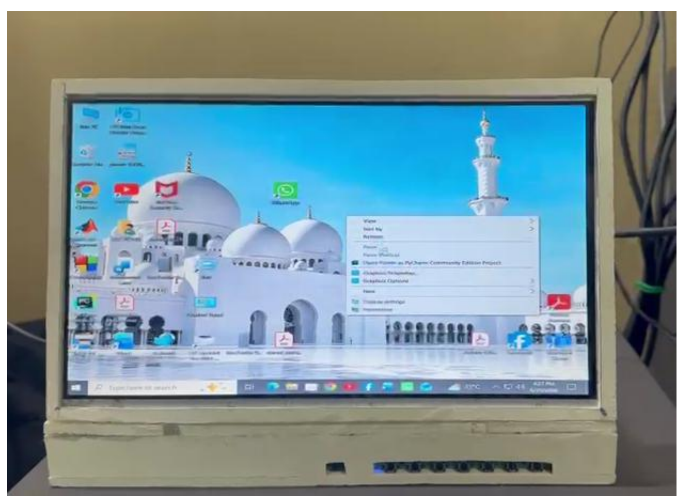

Figure 6: Successful display of the Windows desktop.

8. Problems Encountered and Solutions

Problem	Effect	Solution
Controller compatibility	A controller board matching the LCD panel was not immediately available.	Component compatibility was checked through repeated search and testing before final selection.
Enclosure manufacturing	The intended enclosure exceeded the available 3D-printer build area.	The design was adapted for manual fabrication using PVC board.
Port alignment	The controller-board connectors did not initially align with the enclosure openings.	The openings and internal mounting positions were adjusted during repeated fitting.
Internal cable arrangement	The panel and button cables required careful positioning to avoid strain.	The cables were rerouted and secured before the enclosure was closed.

9. Cost and Feasibility Analysis

The recorded expenditure for the completed prototype was BDT 4,840, including transportation. Excluding transportation, the direct hardware and fabrication cost was BDT 4,040. This indicates that a functional proof-of-concept display can be produced at relatively low cost when compatible components and reusable materials are available.

Parameter	Value
Recorded prototype expenditure	BDT 4,840
Indicative proposed price	BDT 8,000
Gross difference before other expenses	BDT 3,160
Markup on recorded cost	Approximately 65.3%
Gross margin on proposed price	Approximately 39.5%

The figures above are an indicative cost comparison only. They do not include a formal allowance for labour, design time, packaging, warranty, rejected units, taxes, marketing, or after-sales service. A reliable break-even analysis was not calculated because fixed development costs and production quotations were not systematically recorded.

10. Discussion

The project achieved its main objective: a working portable monitor prototype was produced from an LCD panel, a compatible controller board, a power adapter, and a custom PVC-board enclosure. The successful output images confirm that the system accepted a computer signal and generated a readable display. The custom enclosure protected the hardware, provided a viewing support, and kept the controls and connectors accessible.

The strongest aspect of the work was the hands-on hardware and mechanical integration. The project required practical compatibility checking, connector arrangement, measurement, cutting, mounting, and repeated troubleshooting. The manually fabricated enclosure is not as thin or refined as a commercial product, but it demonstrates a functional and repairable proof of concept using low-cost materials.

For reliability, the analysis in this report is limited to observations that were directly demonstrated or recorded during the prototype stage. Unverified specifications and production assumptions have therefore not been used as performance claims.

11. Limitations

- The enclosure was fabricated manually and therefore has visible finishing and alignment limitations.
- The exact panel model, native resolution, refresh rate, weight, and manufacturer specifications were not independently verified in the available records.
- The unit depends on an external power adapter and does not include an internal battery.
- Long-duration operation, temperature rise, electrical safety, drop resistance, and full port-by-port operation were not formally tested.
- The cost analysis is based on one prototype and may not represent production pricing.

12. Future Improvements

- Identify and document the exact LCD panel model and native specifications.
- Use a thinner laser-cut, CNC-machined, or 3D-printed enclosure with improved finishing.
- Add a foldable or adjustable stand and improve the viewing-angle mechanism.
- Improve internal cable management and add strain relief for the panel connector.
- Integrate a rechargeable battery and protected charging circuit if portable power is required.
- Verify HDMI, VGA, AV, USB, audio, and other available interfaces individually.
- Measure power consumption, temperature, weight, brightness, colour performance, and operating duration.
- Add ventilation openings where required after thermal testing.
- Prepare complete circuit/interconnection documentation and verified datasheets for all components.

13. Conclusion

A functional portable monitor prototype was successfully designed, assembled, enclosed, and tested. The author personally completed the key hardware integration and enclosure work, including the external shape, PVC-board fabrication, mounting, connection arrangement, troubleshooting, and final output validation. The prototype displayed a stable computer image and provided usable access to its controls and connectors. The total recorded expenditure was BDT 4,840. Although the enclosure and verification process require further refinement, the project demonstrates practical competence in hardware compatibility, display integration, mechanical fabrication, problem solving, and prototype testing.

Acknowledgements

The author gratefully acknowledges the individuals who provided occasional assistance during component procurement, handling, troubleshooting, and initial documentation support. The core hardware integration, enclosure design and fabrication, assembly, and final testing documented in this report were completed by the author.